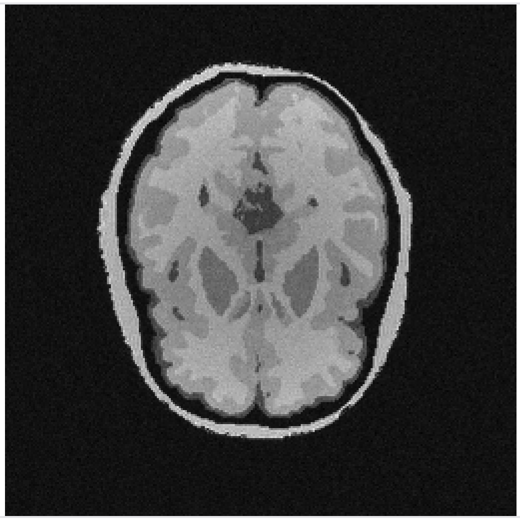
Multimodal CVD (Stroke) Risk Demo

This is a portfolio/research demo, not a clinical tool. Trained entirely on MITRE's synthetic Coherent Dataset (no real patient data, no PHI) with limited compute -- the imaging branch saw only 5 training epochs. Cross-validated performance is modest (AUROC ~0.67-0.68), and the CNN's own Grad-CAM attention often falls outside brain tissue (see notebooks/03_explainability.ipynb). Do not use this for any real medical decision.

MRI brain slice



Age	Sex
85	male

Systolic BP	Diastolic BP
135	47

Total cholesterol	HDL cholesterol
185.8	75.8

BMI	Glucose
27.5	98.3

Smoking status

Never smoker

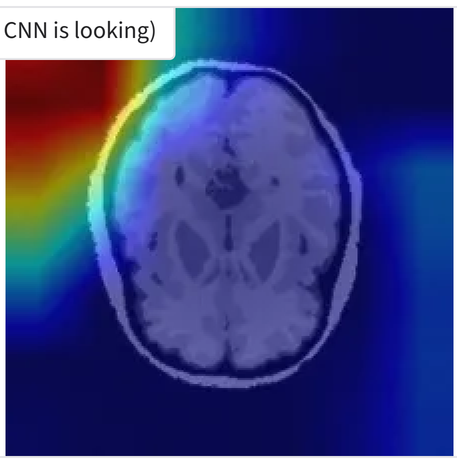
Predict risk

Predicted stroke risk: 49.3%

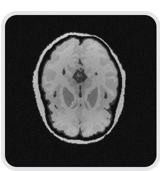
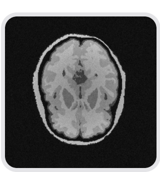
Component	Probability
Imaging model (CNN)	56.3%
EHR model (logistic regression)	44.7%
Blend weight (alpha, imaging share)	0.40

Risk score = alpha x imaging + (1 - alpha) x EHR.

Grad-CAM overlay (where the CNN is looking)



Example patients (real de-identified synthetic cases)

MRI brain slice	Age	Sex	Systolic BP	Diastolic BP	Total cholesterol	HDL c
	82	male	165	85	172.7	
	85	male	135	47	185.8	

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